

2. PROJECT DESCRIPTION

2.1 Project Perspective

MediSync is intended as a replacement for the university clinic's current manual paper based process, and in doing so also serves as an improvement of the process itself rather than only a change in tools. The clinic already carries out registration, consultation, queueing, medicine tracking, and document issuance every day, but each task depends on locked filing cabinets, handwritten logbooks, and a separate Excel file the nurses update by hand. MediSync keeps these same core functions but moves them into a single digital system, so records, schedules, and inventory data exist in one place instead of being spread across paper and separate files, and so that steps such as sorting patients by urgency or generating a monthly report no longer depend entirely on manual work.

Within the university, MediSync will operate as the clinic's own standalone system, separate from other university platforms such as enrollment, though it draws basic student information from the school's existing student information system to avoid encoding the same details twice. The system is built around the understanding that the clinic's responsibility extends to anyone within the campus vicinity, not only enrolled students, so it also supports a manual registration path for faculty, staff, and outside visitors who do not already have a record on file. This direction also supports the Philippine Universal Health Care Law, which calls on primary care providers to move patient records into electronic systems.

MediSync is delivered through two connected surfaces rather than a single one, a web based portal for clinic staff and a mobile application for students, faculty, and staff, both drawing from the same centralized database so a record updated on one surface is immediately reflected on the other. This split reflects how each group actually works, since clinic staff are typically seated at a desk or front counter and benefit from a larger screen, while patients are more likely to check an appointment or their queue status while on the move.

Everyone involved in clinic operations will use MediSync, though access is limited according to role. Students book appointments and check their queue number, employees and outside visitors do the same through their own registration path, nurses record vital signs and update stock, the doctor and dentist issue prescriptions and certificates, and the system administrator manages accounts and security. Together these roles let the clinic continue serving close to a hundred patients a day, now through one organized system instead of a paper based one, and the system is meant to remain the clinic's foundation as it moves from this initial prototype into full implementation during Software Engineering 2.

2.2 Conceptual Framework

MediSync's overall process can be represented using an input process output model, which shows how information supplied by patients and clinic staff moves through the system to produce a usable result. Figure 2.1 presents this conceptual framework.

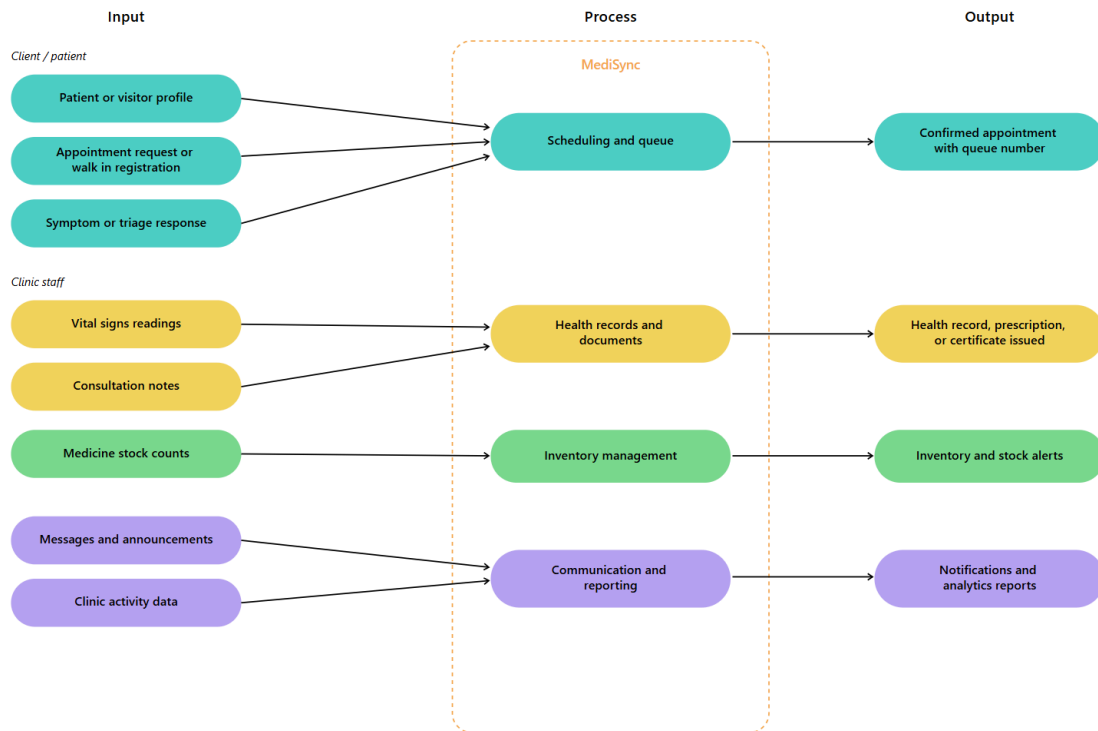


Fig. 2.2.1 Conceptual Framework

Input refers to what enters the system, namely patient profile details, appointment requests, vital signs, triage responses, medicine stock counts, and certificate or prescription requests. Process refers to what MediSync does with that input, based on the modules described in Section 2.4, which include registering patients, scheduling appointments, sorting the queue by urgency, updating health records, tracking medicine stock, and generating prescriptions, certificates, and reports, with access to each function limited by user role. Output refers to what the system produces as a result, namely a digital health record, a confirmed appointment with a queue number, an issued prescription or certificate, inventory reports, a monthly summary report, and analytics for administrators.

This framework shows that MediSync functions as a continuous cycle rather than a set of separate tools. Information gathered during a single patient visit does not stay isolated within one module, since a recorded vital sign can feed into a health record, support a queue priority decision, and later appear in a monthly report, all from the same original input. This connected flow is what allows the system to reduce the repeated encoding and disconnected records that define the clinic's current paper based process.

2.3 User Characteristics

This section identifies the primary users of MediSync in the daily operations of the university clinic. Each user group carries specific tasks, responsibilities, access privileges, and technical backgrounds, and identifying them clearly helps ensure the system meets actual user needs, enforces appropriate security boundaries, and delivers an experience suited to each role across the university community. Table 2.1 summarizes each user type together with their role, responsibilities, access level, and relevant skills.

Table 2.1. User Characteristics

User Type	Role	Main Responsibilities	Level of Access	Relevant Skills
Student	Patient	Books appointments, answers health questions during booking, tracks queue status, and views own past checkup records and excuse slips	Patient level access to own profile, appointments, and health logs	Comfortable with a smartphone or web browser, basic digital skills
Employee or Visitor	Patient	Books or requests a visit, provides basic details for registration if not already on file, answers health questions during intake, and receives care and related documents	Patient level access limited to own profile and visit history, drawn from school records for faculty and staff or entered manually for outside visitors	Basic device use, or willingness to provide information verbally during walk-in registration
Clinic Nurse	Primary Healthcare Staff	Takes and records vital signs, reviews patient health answers, updates daily clinic logs, monitors medicine counts, and assists the doctor and dentist	Staff level access to open patient files, record vital signs, manage the queue, and update medicine stock	Healthcare training, familiarity with daily clinic tasks, basic computer skills for encoding data
University Physician	Clinic Doctor	Examines patients, gives medical advice, issues digital prescriptions, and approves medical certificates	High level access to full patient health history, checkup notes, prescriptions, and certificate approval	Licensed physician, comfortable using standard web based forms
University Dentist	Clinic Dentist	Checks patient teeth, logs dental work, and schedules dental visits and cleanings	High level access to dental histories, treatment notes, and dental scheduling	Licensed dentist, comfortable using basic computer software



Student Volunteer	Clinic Assistant	Helps direct patients to the queue, calls queue numbers, answers simple questions, and assists staff with light clinic tasks	Restricted access limited to public queue numbers and daily staff schedules, no access to private medical files	Good communication skills, familiarity with basic clinic procedures
System Administrator	Technical Support	Creates and manages user accounts and roles, monitors system activity, performs database backups, and resolves technical issues	Full system access to configuration settings, user permissions, security rules, and backups	Background in computer science, knowledge of database management and system security

2.4 Project Components

MediSync is built on several interconnected modules that work together to move the clinic's operations into one digital system. Each module addresses a specific gap in the current paper based workflow, and Table 2.2 summarizes each component together with a short description of its purpose.

Table 2.2. Project Components

Component or Module	Description
Role Based Access and Security Module	Gives each user a secure login and shows only the tools and records that fit their clinic role, keeping private health data safe
Electronic Health Records Module	Stores patient medical histories, past checkups, and vital signs online so staff no longer need to search paper folders
Smart Appointment and Triage Module	Lets patients view open clinic hours, schedule a visit ahead of time, and answer simple symptom questions so urgent cases get seen first
Digital Queue Module	Organizes the waiting line during busy school weeks, gives patients a live queue number, and shows when their turn comes up
Medicine Inventory Tracker Module	Tracks clinic supplies automatically and records the start, dispensed, restock, and end count of each medicine daily, using both generic and brand name, sends low stock warnings, and prepares the monthly stock summary
Automated Prescription Module	Connects consultations with the clinic medicine shelf, letting the doctor write an order that updates the patient file and lowers stock count in one step
Medical Certificate Generator Module	Builds official printable medical certificates directly from consultation notes so staff do not have to write them by hand
Communication Module	Lets clinic staff send messages or announcements to students, staff, or parents about appointments, follow ups, or general clinic notices, delivered through in system notifications and the university Outlook email
Health Analytics Dashboard Module	Gathers clinic numbers to display total visits, busy hours, common sicknesses, and medicine use in easy to read charts, and prepares the monthly summary report of consultations and medicines released
Health Engagement Module	Presents basic health information through a simple explorable campus map paired with short quizzes, encouraging engagement with health topics outside of an actual clinic visit

2.5 Project Constraints

MediSync comes with several constraints that could influence its development and later operation. Since the system is being built within a single academic term by a group of students rather than a professional team, the schedule is fixed by the course calendar and the team's working knowledge of Python, web and mobile development, database management, and machine learning is limited to what has been learned and practiced during the project, with no clinical experts on hand beyond the clinic personnel and adviser who provide feedback on workflows. The project also works with a small budget, so development tools, frameworks, and libraries are mainly free and open source, which affects SMS notifications in particular since gateways typically charge per message and testing may only be possible at a small scale.

On the technical side, MediSync needs a server and database for patient records, appointments, queue data, inventory, prescriptions, and certificates, along with a Python environment and its required frameworks, reached through a supported web browser for the clinic staff portal and through the mobile application for patients, though actual performance under the clinic's real setup can only be confirmed after deployment since development is limited to the equipment the team already has. The system is designed to run on the university's local area network, so both the portal and the mobile app can only be used within the campus and depend on that connection's stability, with no offline access in the current version, meaning clinic operations may be interrupted if the network goes down.

Since the Electronic Health Records, Digital Queue, and Medicine Inventory Tracker modules depend on accurate data, existing patient information, schedules, and medicine records must be reviewed and encoded before the system can be used properly, and because real health records are confidential, the team can only test with sample data rather than verifying how the system handles actual records before deployment. The machine learning used for priority classification is similarly limited by the quality and amount of available data, so its low, medium, and high priority levels are meant only to help organize the queue rather than diagnose a condition or replace clinic judgment, and the system is not meant for critical emergencies, which still require immediate in person attention. Finally, since MediSync handles health related information, it must follow university policy and Philippine data privacy requirements, which limits how much data, including analytics, can be shared with school leaders, and SMS notifications depend on an outside gateway whose delivery can be affected by provider availability, cost, and the accuracy of registered numbers.

2.6 User Interface

MediSync reaches its users through two connected surfaces, a web based portal for clinic staff accessed through a browser on a computer or other compatible device, and a mobile application that students, faculty, and staff use as patients to book visits, check their queue status, and view their own records. Both surfaces connect to the same university local network and the same centralized database, so records stay consistent no matter which surface a person uses. All users pass through a single login page, after which the pages and tools they see depend on their assigned role, keeping the interface simple for each user and keeping private records away from those not authorized to view them.

For students, the interface opens on a home page showing their upcoming appointment and current queue status, from where they can book a visit against the available doctor and dentist schedules, answer a short symptom questionnaire, check their live queue number, view a profile page limited to the parts of their record they are allowed to see, such as past checkup notes, excuse letters, and medical passes, read messages sent by clinic staff, and open the health engagement map and quiz. Employees and outside visitors follow a similar but simpler path, using a booking or registration page to request a visit or complete walk in registration, then checking their own queue number and a visit history limited to their own records.

For nurses, the interface centers on a queue management screen that includes both the priority queue and a separate queue for busy days, alongside patient record pages for recording vital signs, a health checkup form, a daily clinic log sheet, a medicine inventory page with low stock alerts, an incident report form, and a monthly summary report screen. Doctors work from a list of patients waiting to be seen arranged by priority, opening patient health history pages, checkup notes, and prescription screens, and generating excuse letters, medical passes, medical certificates, or a go home form as each case requires. Dentists follow a comparable path suited to dental care, working from their own list of waiting patients through dental history pages, a dentist chart, treatment notes, a referral form, and the dental appointment schedule.

Student volunteers see only a public queue number display and the daily staff schedule, along with their own volunteer agreement page, since their access does not extend to private medical files. System administrators work through a user account and role management page, system settings, activity logs, and a database backup page. Clinic staff authorized to view reports can also open the health analytics dashboard, which shows visit counts, busy hours, common illnesses, and medicine usage in chart form, and can prepare the monthly summary report from the same dashboard. Clinic staff can also open a messaging screen to send a message directly to a student or a parent regarding an appointment or a concern, separate from the automatic notifications the system sends on its own.



Navigation follows a menu that stays visible on the screen, such as a sidebar on the staff portal or a bottom menu on the mobile app, and only lists the modules the logged in user is allowed to use, so a student will not see inventory or prescription options, for example. Each module has its own page, users can return to their home page from any screen, and common actions such as booking an appointment or opening a patient record are placed within one or two taps or clicks.

Interaction with the system happens mainly through forms, tables, buttons, and search fields. Patients fill out simple forms when booking an appointment or answering symptom questions, while clinic personnel enter vital signs, consultation notes, and stock updates through forms and use search fields to find a patient record or medicine item. The queue displays as a list of numbers that updates as patients are called, and the system shows notifications for appointments, queue updates, and low stock warnings both within the application and through the university Outlook email, while SMS messages go to the mobile numbers registered by users. Actions that change or remove records, such as deleting an entry or finalizing a prescription, ask for confirmation first to prevent mistakes.

Usability is aimed at users with different levels of comfort with technology, from students comfortable with mobile apps to clinic personnel who mostly work with paper records. Screens use plain labels, a consistent layout, and short forms with only the fields needed, and important information such as queue numbers, priority levels, and low stock warnings is made easy to notice. Since the clinic gets busy during peak school weeks, common tasks such as updating the queue or recording vital signs are kept to as few steps as possible, error messages are written in simple language, and the system logs users out after a period of inactivity to protect private health data on shared clinic computers.

2.7 Assumptions and Dependencies

MediSync's operation rests on several assumptions about the clinic, its users, and its technical environment. The system is expected to be reached mainly through a web browser on a computer or other compatible device, with the mobile application serving as an additional access point for students and staff, and both are assumed to run within a stable university local area network that can support authorized access to MediSync and its centralized database throughout clinic hours without frequent interruption. It is also assumed that the clinic will have functioning ID readers in place at its point of entry, since the tap in check in feature that confirms a patient's arrival and matches it to a booked appointment depends entirely on that hardware working consistently.

Students, faculty, staff, and other university personnel are expected to be the primary users who book appointments, receive care, and view the portions of their own records that the system allows, while clinic personnel manage records, consultations, appointments, queues, inventory, prescriptions, certificates, and reports according to their assigned roles. Outside visitors within the campus vicinity are assumed to provide accurate information during manual walk in registration, since their records do not already exist in the system the way an enrolled member's does, and clinic staff are assumed to be trained not only on the system itself but specifically on this manual registration process so that visitors are served without confusion or delay. Clinic personnel more broadly are assumed to correctly encode and maintain accurate, up to date information across every module, since the reliability of the Electronic Health Records, Digital Queue, and Medicine Inventory Tracker modules depends entirely on the accuracy of what staff enter.

Every patient, whether an enrolled member or a walk in visitor, is assumed to read and accept the terms and conditions and consent notice presented at registration or first login before their information is collected, and this consent is assumed to be sufficient under university policy and Philippine data privacy requirements for the system to store and use their data for clinic purposes. The priority queue, which classifies cases as low, medium, or high, is assumed to support non critical clinic operations rather than replace professional judgment, so it does not diagnose a patient's condition and is not intended for critical emergencies or other situations that need immediate medical attention, which are assumed to proceed directly to the clinic or appropriate emergency care instead of through the system. Role based access is assumed to be enforced consistently across every module, so each user holds an account and permissions matching their responsibilities, sensitive functions stay limited to authorized personnel, and patients can view appropriate portions of their own records while being kept out of records belonging to others.

The health engagement feature is assumed to be used voluntarily by students as a supplementary source of basic health information, and its quizzes are assumed never to substitute for an actual consultation, diagnosis, or professional medical advice from clinic staff. Notifications sent through the

in system inbox and the university Outlook email are assumed to reach students and staff reliably, which in turn assumes that the contact details on file, whether pulled from the school's existing student information system or encoded manually for visitors, remain accurate and current.

Beyond these assumptions, MediSync depends on a properly configured server and database for storing all of its records, on the required Python runtime, frameworks, and libraries that support both the application and its machine learning functionality, and on sufficient and well prepared data, algorithms, and regular validation to keep that machine learning useful for queue prioritization. Its notification feature depends on both an SMS gateway capable of delivering messages to registered numbers, which can be affected by provider availability, limitations, and cost, and on the university's existing Microsoft 365 and Outlook infrastructure, which the system relies on to deliver email based notifications without needing to build and maintain a separate mail service. The tap in check in and appointment features depend on functioning RFID or similar card reading hardware installed at the clinic and on clinic personnel keeping doctor and dentist schedules accurate, while the inventory functions depend on personnel keeping medicine records accurate and current.

MediSync also depends on the university's existing student information system remaining accessible enough to supply the basic student details the system pulls from it, since any change to that system or to the university's willingness to share this data could affect how much information MediSync can retrieve automatically versus what staff must encode by hand. Continued operation depends on regular maintenance, database backups, security updates, and troubleshooting support, on the ongoing participation of clinic personnel and administrators who provide requirements, validate workflows, and use the system day to day, and on sustained institutional support from the university, since a system built by a student team for an academic project ultimately depends on the school choosing to adopt, fund, and maintain it beyond the course itself. Finally, the system depends on continued compliance with university policy and Philippine requirements concerning privacy, electronic records, and the handling of health related information, since any change in these requirements could call for adjustments to how MediSync stores, shares, or restricts access to patient data.

2.8 Proposed Future Requirements

The continuous development of MediSync opens room for future enhancements that go beyond its current required version. These planned features are not part of the system's present scope, but they represent directions the team believes are worth pursuing as the project moves from this Software Engineering 1 prototype into full implementation during Software Engineering 2 and beyond.

2.8.1 Advanced Machine Learning Features

Future versions may build on the machine learning already used for queue priority classification by adding models that predict clinic demand during specific periods such as enrollment or exam weeks, identify recurring patterns in medicine consumption to anticipate restocking needs earlier, forecast overall inventory requirements across a school term, and analyze appointment trends to flag illnesses that appear to be spreading among students. These additional capabilities would remain decision support tools that inform clinic staff rather than replacements for their own clinical judgment.

2.8.2 AI-Assisted Symptom Guidance

A future version may introduce a conversational assistant that helps a student or visitor describe their symptoms in plain language before a consultation, offering general guidance on what to expect and helping the system suggest a more accurate priority level. Any such feature would need to state clearly that it does not diagnose conditions and that a clinic staff member always makes the final call on priority and treatment.

2.8.3 Deeper University System Integration

Beyond the basic student information already pulled from the school's existing system, MediSync may eventually connect more fully with enrollment, faculty, or employee databases to further reduce duplicate data entry across the university, and could extend this integration to automatically flag when a student's enrollment status changes, such as a leave of absence or graduation, so their clinic record is updated accordingly without manual follow up.

2.8.4 Online Medical Clearance Processing

Students may eventually initiate and complete more parts of the medical clearance process online before visiting the clinic, such as submitting a health declaration or uploading prior medical documents in advance, subject to the clinic's own procedures and requirements, which would shorten the in person portion of a clearance visit to mainly the parts that genuinely need a nurse or doctor present.

2.8.5 Automated Inventory Replenishment

The inventory module may eventually generate automatic replenishment recommendations or purchase requests once medicine and supply levels reach predefined thresholds, and could extend further into suggesting order quantities based on past consumption patterns, reducing the manual work involved in deciding how much of each item to reorder.

2.8.6 Advanced Analytics Dashboard

Future versions may provide dashboards with historical trends, peak periods, visit patterns, medicine utilization, and other operational indicators beyond what the current dashboard shows, potentially allowing school leaders to compare data across different terms or academic years to support longer term planning for clinic staffing and budget.

2.8.7 Digital Document Verification

Certificates, prescriptions, and other clinic documents may eventually carry electronic signatures, QR codes, or other verification methods that let a receiving office, such as a professor or an employer, confirm a document's authenticity directly rather than relying on its printed appearance alone.

2.8.8 Parent or Guardian Access for Minors

A future version may give parents or guardians of senior high school students controlled notification or access to relevant clinic information, subject to authorization and privacy requirements, which would extend the release process already in place for minors into a more proactive channel where a parent is notified automatically rather than only being contacted when their child is ready to be picked up.

2.8.9 Offline Mode and Data Synchronization

MediSync may eventually support limited offline functionality for selected operations during network interruptions, such as recording a vital sign or a walk in registration locally before the connection returns, with those records synchronized automatically once connectivity is restored, reducing the disruption a network outage would otherwise cause during a busy clinic day.

2.8.10 Integration with Medical Devices

Future versions may support compatible devices, such as digital thermometers, blood pressure monitors, or pulse oximeters, that automatically transfer their readings into a patient's electronic health record, reducing the manual entry of vital signs and lowering the chance of a nurse mistyping a reading during a busy period.

2.8.11 External Referral and Emergency Recording

Future development may add features for documenting referrals to external healthcare facilities when a case exceeds what the clinic can handle, and for recording emergency related



incidents that clinic staff respond to. Even though the system itself is not meant to manage an emergency as it happens, keeping a record afterward would still support the clinic's own reporting and help identify patterns in the kinds of cases being referred out.

2.8.12 Expanded Health Engagement Content

The health engagement module may grow beyond its initial quiz and map format to include a wider range of health topics, seasonal content tied to concerns such as flu season or exam stress, and simple recognition such as badges for students who complete a set number of health checkups or quizzes, building further on the gamification direction the university already encourages for student projects.
